

SANTANDER, Spain

WMO: 080230

Lat: 43.4911N

Lon: 3.8006W

Elev: 59

StdP: 100.62

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	4.0	4.9	-1.8	3.3	7.1	-0.3	3.7	7.5	17.3	9.3	15.5	10.1	4.5	150	0.478

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.6	25.4	19.6	24.1	19.7	23.1	19.4	21.7	23.7	21.0	22.9	20.3	22.2	5.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.0	15.8	23.0	20.2	15.0	22.2	19.6	14.4	21.7	63.5	23.8	60.9	23.0	58.7	22.3	24.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.8	11.9	10.3	2.1	30.1	1.5	2.3	1.1	31.8	0.2	33.1	-0.6	34.4	-1.6	36.0		
(5)			0.6	23.3	1.6	0.9	-0.5	23.9	-1.5	24.4	-2.4	24.9	-3.6	25.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	14.8	10.7	10.4	11.7	12.7	14.8	17.5	19.6	20.2	18.7	16.9	13.2	11.4	(6)
(7)		DBStd	4.21	2.74	2.81	2.79	2.21	2.18	1.86	1.64	1.54	1.92	2.83	2.72	2.91	(7)
(8)		HDD10.0	88	24	26	12	3	0	0	0	0	0	0	4	18	(8)
(9)		HDD18.3	1441	238	223	206	171	111	36	7	2	18	62	156	214	(9)
(10)		CDD10.0	1856	44	36	66	83	150	226	296	318	260	214	100	63	(10)
(11)		CDD18.3	168	0	0	1	1	2	12	44	61	28	17	2	0	(11)
(12)		CDH23.3	222	0	0	2	2	12	25	38	59	36	47	1	0	(12)
(13)		CDH26.7	29	0	0	0	0	2	7	3	8	4	4	0	0	(13)
(14)	Wind	WSAvg	4.5	5.2	5.1	5.0	4.6	4.3	3.9	3.7	3.7	3.9	4.3	5.3	5.1	(14)
(15)	Precipitation	PrecAvg	1128	106	86	84	96	70	60	44	69	95	101	162	131	(15)
(16)		PrecMax	1472	239	235	151	237	153	240	102	204	256	296	436	367	(16)
(17)		PrecMin	750	5	8	0	21	21	12	13	13	16	12	35	9	(17)
(18)		PrecStd	174	56	52	39	60	34	44	25	50	60	75	83	80	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.1	19.9	22.6	22.1	24.7	26.6	26.5	26.8	26.7	26.9	22.3	19.7	(19)
(20)			MCWB	12.5	12.1	14.4	14.8	16.6	20.2	21.8	21.5	18.8	18.2	16.0	13.1	(20)
(21)		2%	DB	17.3	17.6	19.6	19.1	20.7	22.5	24.1	24.6	24.0	24.5	20.2	18.2	(21)
(22)			MCWB	11.9	11.6	13.4	13.7	15.9	18.9	21.0	21.0	19.5	17.2	14.7	12.6	(22)
(23)		5%	DB	15.9	15.7	17.7	17.2	19.1	21.2	23.1	23.6	22.7	22.8	18.6	16.8	(23)
(24)			MCWB	11.5	11.3	12.3	13.4	15.9	18.3	20.4	20.5	19.2	16.7	14.1	12.0	(24)
(25)		10%	DB	14.6	14.2	15.7	15.8	17.9	20.2	22.3	22.8	21.6	21.1	17.1	15.5	(25)
(26)			MCWB	11.0	10.8	11.9	13.2	15.3	17.8	19.7	20.0	18.9	16.6	13.7	11.6	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.0	13.9	15.2	16.2	18.4	20.9	22.7	23.1	21.7	19.8	17.3	14.7	(27)
(28)			MCDWB	16.5	17.3	20.0	19.2	21.7	24.1	25.0	25.8	23.8	22.6	20.0	17.1	(28)
(29)		2%	WB	13.0	12.8	14.1	14.9	17.2	19.6	21.6	21.8	20.7	18.9	16.0	13.7	(29)
(30)			MCDWB	15.2	15.4	17.8	17.4	19.5	21.7	23.5	23.7	22.6	21.6	18.3	16.2	(30)
(31)		5%	WB	12.3	12.1	13.2	14.1	16.4	18.8	20.8	21.1	20.1	18.1	15.1	13.0	(31)
(32)			MCDWB	14.5	14.3	16.3	16.2	18.4	20.5	22.7	23.0	21.7	20.8	17.3	15.4	(32)
(33)		10%	WB	11.7	11.4	12.5	13.5	15.7	18.2	20.1	20.5	19.3	17.4	14.3	12.3	(33)
(34)			MCDWB	13.8	13.5	15.1	15.4	17.4	19.8	21.9	22.3	21.0	20.0	16.5	14.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.1	4.5	4.8	4.8	4.4	4.2	4.2	4.6	4.8	5.1	4.3	4.2	(35)
(36)			MCDBR	5.7	6.6	7.9	7.3	6.7	5.8	5.5	5.8	6.3	7.6	6.3	5.7	(36)
(37)			MCWBR	3.5	4.0	4.2	4.2	3.8	3.6	3.6	3.6	3.9	3.8	3.6	3.5	(37)
(38)		5% WB	MCDBR	4.7	5.3	6.1	6.0	5.5	4.9	5.2	5.2	5.3	5.8	5.0	4.9	(38)
(39)			MCWBR	3.6	3.8	3.9	4.0	3.6	3.4	3.5	3.5	3.6	3.7	3.6	3.7	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.331	0.343	0.377	0.397	0.407	0.419	0.407	0.404	0.385	0.366	0.344	0.326	(40)	
(41)		taud	2.497	2.449	2.356	2.311	2.317	2.318	2.353	2.372	2.419	2.475	2.485	2.503	(41)	
(42)		Ebn at Noon	799	850	859	868	866	855	860	852	841	813	773	768	(42)	
(43)		Edh at Noon	72	88	109	123	126	126	120	115	101	85	72	66	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.38	2.16	3.38	4.53	5.35	5.74	5.76	5.14	4.17	2.73	1.59	1.28	(44)	
(45)		RadStd	0.16	0.29	0.40	0.44	0.36	0.42	0.27	0.31	0.18	0.22	0.19	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (9 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page